5 Findings

In this chapter the findings from both phases, quantitative survey and qualitative interviews, are presented. The quantitative survey results establishes how the UTAUT constructs and Self-Efficacy relate to LLM usage frequency among public administration employees. The qualitative interview findings then explain the mechanisms behind the observed correlations, particularly the dominance of Self-Efficacy and the weakness of Facilitating Conditions. The chapter proceeds as follows. Section describes the sample characteristics. Section reports construct reliability, descriptive statistics and bivariate correlations. Section 4.1.3 presents the regression analysis. Section 4.1.4 examines task application patterns. Section 4.1.5 identifies adoption barriers among non-users.

Quantitative Findings

Sample Characteristics

The final sample consisted of n=189 respondents. 34 responses were excluded either due to incomplete data or non-eligibility (people were asked whether they work in German public administrations or not), so that after data cleaning a final analytical sample of n=154 remained. Throughout the rest of this thesis, “sample” refers to this analytical sample of n=154, unless otherwise noted.

Table X summarized the demographic characteristics of the sample.

Age Group	Entire Sample	% of Total
Unter 30 Jahre	16	10.39%
30 - 39 Jahre	48	31.17%
40 - 49 Jahre	27	17.53%
50 - 59 Jahre	22	14.29%
60 Jahre oder älter	16	10.39%
k. A.	25	16.23%
Total	**154**	**-**

The age distribution in the sample showed the largest group 30-39 age group (31.17%, n=48), followed by 40 - 49 age group (17.53%, n=27), 50 - 59 age group (14.29%, n=22), over 60 age group (10.39%, n=16) and under 30 age group (10.39% , n=16). A portion of respondents (16.23%, n=25) did not disclose their age in the survey.

The sample captured the entire landscape of German public administrations.

Agency Type	**Entire Sample**	**% of Total**
Bundesbehörde	19	12.34%
Landesbehörde	28	18.18%
Kommunale Verwaltung (z. B. Stadt, Landkreis, Gemeinde)	31	20.13%
Sonstige Einrichtung des öffentlichen Rechts	26	16.88%
University & Research	16	10.39%
Judiciary	2	1.30%
Church/Religious Body	2	1.30%
k. A.	30	19.48%
Total	**154**	**-**

Municipal administrations were being most represented (20.13%, n=31), followed by other public law institutions or respondents who did not specify their organizational affiliation (19.48%, n=30), state authorities (18.18%, n=28) and federal authorities (12.34%, n=19). Universities and research institutions accounted for 10.39%, n=16) of the sample, while the judiciary made up 1.3% (n=2).

Of these n=154, 116 (75.32%) reported having used LLMs for work-related tasks, while 38 (24.68%) reported no prior work-related LLM use or having used it for testing purposes.

Table X shows the types of LLM tools that respondents of the user subsample indicated to use primarily.

Category	**n**	**% of Total**
Freely accessible	97	83.60%
Org-provided	16	13.80%
Not classifiable	3	2.60%

The majority relied on public available tools (n=97, 83.6%), such as ChatGPT, Perplexity or Claude. Only 13.8% (n=16) reported using institutionally provided tools such as Microsoft Copilot, LLMoin or NRW.Genius. There were 3 tools (2.6%) that were not classifiable. (ChatGPT was by far the most frequently reported individual tool (65,5% of users) (see

Appendix X for the entire breakdown). This tool distribution is consistent with what was reported on the access landscape in Section Error: Reference source not found.

Regarding the usage frequency please refer to Table X.

Age Group	**Nein / Nie**	**Seltener <1×/Mo**	**Mehrmals/ Mo**	**≈1×/ Woche**	**Mehrmals/ Woche**	**n**
Unter 30 Jahre	6	1	1	2	6	**16**
30 - 39 Jahre	7	4	9	11	17	**48**
40 - 49 Jahre	10	4	1	4	8	**27**
50 - 59 Jahre	4	7	6	2	3	**22**
60 Jahre o. älter	5	5	0	0	6	**16**
Sum	38	21	17	19	40	-
In %	24.81%	16.28%	13.18%	14.73%	31.01%	-

The most common usage frequency was multiple times per week (n = 40, 48), which indicates that the user subsample was characterized by more heavy users. This may reflect the convenience sampling strategy, which may have overrepresented employees with existing interest in LLMs. Usage frequency varied across age groups, with the 30-39 age group reporting the highest intensity. All of the following construct analyses are based on the user subsample (n=116), because the UTAUT constructs and Self Efficacy items were only relevant for respondents who actually made use of LLMs in the past.

Construct Reliability, Descriptives and Correlations

Before examining the construct scores, internal consistency was assessed for each composite score using Cronbach's alpha (see Table X). Cronbach's alpha measures how closely related a set of items are as a group. Performance Expectancy was consistent (α = 0.72). Facilitating Conditions (α = 0.64) and Self-Efficacy (α = 0.60) were within the range considered tolerable for exploratory research with few items. Effort Expectancy (α = 0.52) and Social Influence (α = 0.41) showed low internal consistency. However, mean inter-item correlations for all constructs were with the recommended range of 0.15 to 0.50 for scales with few items (Source), which means that the items within each construct are meaningfully related to each other despite the subpar alpha values (see Table X). Nonetheless, the findings concerning Effort Expectancy and Social Influence should be interpreted with caution.

Table X shows the descriptive statistics for the four UTAUT predictor constructs, Self-Efficacy as well as the dependent variable usage frequency.

Construct	M	SD	α	Mean r
Performance Expectancy	2.68	0.65	0.72	0.49
Effort Expectancy	2.44	0.71	0.52	0.26
Social Influence	3.34	0.73	0.41	0.20
Facilitating Conditions	2.76	0.84	0.64	0.39
Self Efficacy	3.77	0.85	0.60	0.36
Usage Frequency (Dependent Variable)	2.81	1.20	-	-

For each construct, a composite score was calculated as the mean of the respective items for the LLM user group (n=116). As already indicated in 4.2.2, the items were measured on 5-point scale, which means that the composite scores range from 1 to 5. Usage frequency was measured on a 4-point scale ranging from "less than once a month" coded as 1 to "multiple times per week" coded as 4. The original survey item included a fifth option ('never used") which was used to distinguish users from non-users. Only the four active usage categories are included here. Only respondents who reported having used LLMs for work (n = 116) are included. Self -Efficacy had the highest mean score (Mean = 3.77, Standard Deviation = 0.85). On a 5-point scale, this indicates that respondents on average reported confidence above the scale midpoint in their ability to evaluate and work with LLM outputs.

As a preliminary step before the regression analysis in Section 5.3, Table X presents the bivariate Pearson's r correlations between each construct score and usage frequency. This provides an initial overview of which constructs are associated with LLM usage frequency before testing whether those associations hold up in the regression.

Construct	r	p
Performance Expectancy	0.36	< 0.001
Effort Expectancy	0.14	0.142
Social Influence	0.38	0.000
Facilitating Conditions	0.13	0.201
Self Efficacy	0.44	< 0.001

All constructs showed positive correlations with usage frequency. Self-Efficacy showed the strongest bivariate correlation ($r = 0.44$, $p < 0.001$). Social Influence ($r = 0.38$, $p <$

0.001) and Performance Expectancy (r = 0.36, $p < 0.001$) also reached significance. Facilitating Conditions (r = 0.13, p = 0.201) and Effort Expectancy (r = 0.14, p = 0.142) did not reach statistical significance. These bivariate relationships provide initial context but do not account for shared variance among predictors. To address, the regression analysis is presented in the next Section.

Regression Analysis

To determine which constructs independently predict LLM usage frequency when all five are modeled simultaneously, an OLS multiple regression was conducted. Table X presents the full regression results.

Construct Name	β	p	VIF
Performance Expectancy	0.24	0.016	1.32
Effort Expectancy	0.12	0.150	1.06
Social Influence	0.19	0.069	1.50
Self Efficacy	0.29	0.01	1.67
Facilitating Conditions	-0.1	0.315	1.35

When all five constructs were entered simultaneously as predictors of usage frequency, the model explained of the variation in usage frequency ($R^2 = 0.29$). VIF for all predictors was below 5, which confirms that the five constructs were distinct enough for the regression to reliably separate their individual effects.

Two constructs were significant independent predictors. Self-Efficacy was the strongest (β = 0.29, p = 0.01), indicating that confidence in evaluating LLM outputs predicted more frequent use even after accounting for the other four constructs. Performance Expectancy was the second significant predictor (β = 0.24, p = 0.02), which means that perceived usefulness of LLMs also independently predicted usage frequency.

Social Influence did not reach statistical significance but came close (β = 0.19, p = 0.07). Effort Expectancy (β = 0.13, p = 0.15) and Facilitating Conditions (β = -0.10, p = 0.32) did not reach statistical significance in the regression. Notably, Facilitating Conditions showed a positive bivariate correlation with usage frequency (r = 0.13) but a slightly negative value in the regression (β = -0.10). In other words, when FC was considered on its own, it appeared to be associated with more frequent LLM use. But when the other four constructs were accounted for, this connection went away. This may suggest that FC's bivariate correlation was not reflecting its own relationship with usage frequency but rather its overlap with other constructs that do independently predict usage.

Adoption Barriers Among Non-Users

As a final step in the quantitative analysis, the barriers reported by the non-user subsample were examined. As reported in Section 5.1.1, 38 of the n=154 respondents (24.68%) indicated that they had not used LLMs for work-related tasks. These non-users were asked which factors had kept them from using LLMs at work. Multiple selections were possible, so the reported percentages do not sum to 100%. Table X presents the barriers in order of how frequently they were named.

Barrier	**n**	**% of Non-Users**
Fehlende Schulung/Guidelines	16	42.11%
Kein Mehrwert/Qualitätszweifel	15	39.47%
Keine Freigabe/Zugang	14	36.84%
Informationssicherheitsbedenken	13	34.21%
Unklare Rechts-/Datenschutzlage	11	28.95%
Fehlende Integration	11	28.95%
Verbot durch Dienstvereinbarung	7	18.42%
Sonstiges	6	15.79%
Zeitmangel zur Einarbeitung	5	13.16%

The most frequently named barrier was a lack of training or guidelines (Fehlende Schulung/Guidelines), reported by 42.1% (n=16) of non-users. This was followed closely by doubts about the added value or output quality of LLMs (Kein Mehrwert/Qualitätszweifel, 39.5%, n=15) and the absence of official clearance or access (Keine Freigabe/Zugang, 36.8%, n=14). Information-security concerns (Informationssicherheitsbedenken) ranked fourth at 34.2% (n=13). An unclear legal or data-protection situation (Unklare Rechts-/Datenschutzlage) and a lack of integration into existing workflows (Fehlende Integration) were each named by 28.9% (n=11). The remaining barriers were reported less often: a prohibition through a service agreement (Verbot durch Dienstvereinbarung, 18.4%, n=7), other reasons (Sonstiges, 15.8%, n=6) and a lack of time to become familiar with the tool (Zeitmangel zur Einarbeitung, 13.2%, n=5).

Most of the barriers reported most frequently relate to the organizational conditions surrounding LLM use rather than to the tool itself. Missing training and guidelines, the absence of official access, a lack of workflow integration, information-security concerns and an unclear legal or data-protection situation each point to a form of organizational provision, clearance or governance that non-users experienced as absent, unclear or inadequate. The main exception among the most frequently named barriers was doubt about the added value or quality of LLM output (Kein Mehrwert/Qualitätszweifel), which concerns the perceived usefulness of the tool rather than the conditions surrounding its use.

As the non-user subsample is small (n=38) and respondents could name more than one barrier, these figures are reported as descriptive and are not used for statistical comparison. They serve to characterize the obstacles that non-users themselves identified.

An item on how clearly organizations had communicated rules for LLM use was added during data collection and was therefore answered only by a subsample of 18 of the 116 users. Because it was introduced later, the following figures are reported as supplementary and descriptive. Of these 18 users, 14 reported that their organization had communicated no rules for LLM use, or that they were not aware of any. All 18 reported using the consumer version of ChatGPT. Given the small subsample, these figures are not used for statistical comparison. Rather, they are reported because they characterize the conditions under which users in this sample operated.

Qualitative Findings

This chapter examines how employees came to use LLMs largely on their own and what that use depends on. It should be noted that the majority of the organizations did not provide tools or training.

Three themes emerged from the interviews. First, users rely on their professional knowledge to evaluate LLM output, while non-users name the absence of that knowledge as the central risk. Second, LLM competence was acquired through self-directed experimentation rather than organizational training, often beginning in private contexts before carrying over to work. Third, users decide task by task whether to deploy or not LLMs based on their own judgment, while non-users assess suitability on broader terms. The rest of the chapter will examine these three themes in detail.

Professional Knowledge as the Basis for Evaluating LLM Output

When asked how they proceed when evaluating LLM outputs, users did not point to any formal methodology or checklist. Rather, they described an intuitive process that rests on their professional knowledge. IP03, who works in a technically oriented role, put this vividly. When asked how they checked an output they said, “I think gut feeling. The gut feeling is also that you are implicitly checking your own prior, comparing your own knowledge against what you see there.” (Interview IP03)[1] This gut feeling is not random. As the user described it, it is rooted in one’s own professional knowledge and lets them quickly detect whether the output and what they already know are aligned or misaligned. IP04 described a similar approach with the metaphor of a warning light. When they feel that the model fails to grasp the professional concepts they are working on, this is a warning sign, “One indicator is, for example, whether it understands the terms or concepts that I enter. If I have the feeling that something completely different comes back, that it does not fit the concept at all, then that is a kind of warning light.” (Interview IP04)[2] Their verification here seems to be an internal and experience based judgment rather than a procedural one, a check against the user’s own professional knowledge rather than against a codified reference.

The error signals that set off this intuitive could be concretely named by the users. One signal is when the model produces hallucinations or absurd data. IP03 named this directly, “One warning light is completely absurd data, so hallucinations.” (Interview IP03)[3] IP04 described the same kind of signal as output that is plainly bad and recognizable at once, “With AI answers, when they are bad, you also see it immediately, you also think, wow, that is really quite a lot of slop that the AI model has produced there.” (Interview IP04)[4].

Alongside this intuitive check, users also described a more deliberate verification mode. They engage in it when the stakes of the task are high or when the output runs against

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[1] „Ich glaub Bauchgefühl. Das Bauchgefühl ist ja auch, dass man implizit so seinen eigenen Prior abfragt, die sein eigenes Wissen abgleicht gegen das, was man da sieht.“
[2] „Ein Indikator ist zum Beispiel, ob es die Begriffe oder die Konzepte versteht, die ich rein gebe. Wenn ich das Gefühl habe, da kommt was ganz anderes zurück, das passt überhaupt nicht zu dem Konzept, dann das ist so eine Warnleuchte.“
[3] „Eine Warnleuchte ist ganz absurde Daten – also Halluzinationen.“
[4] „Bei KI-Antworten, wenn die schlecht sind, siehst du es auch sofort, denkst du auch so, wow, das ist jetzt aber ziemlich hier, ziemlicher Slop, dass das KI-Modell da produziert hat.“

what they expected. IP02 reported that for important outgoing documents, especially ones with numbers, they verify rigorously and carry that practice over to AI, "When it comes to numbers, audit reports for example, […] I also check myself double and triple and a colleague checks me again before anything goes out. And that then also carries over to the use of AI." (Interview IP02)[5] IP05 reported a comparable practice when producing project reports, where they first check whether the facts are correct, "With the project reports, I first of all check whether the facts are correct." (Interview IP05)[6] More broadly, IP05 described looking more closely whenever the work carries weight or the result is surprising, "If it is really important or the result surprises me, then of course I check again." (Interview IP05)[7] This deliberate checking mode is not random either. This deliberate checking is based on their professional knowledge, since the user knows which facts or references most likely need checking. Someone without that knowledge would likely not know which parts of an output to scrutinize and check.

By contrast, the non-user accounts confirm the important role of professional knowledge from the opposite direction. They name that not having the relevant checkable knowledge is a risk, when using LLMs. NU02 specifically pointed to working in areas where they lack the background to assess the output, "If it moves into an area where I do not have background knowledge and cannot assess it, I have to trust that what it gives me there is correct. And there I see a bit of a danger that I might be building on false foundations." (Interview NU02)[8] NU03 made the same point from the side of what they already know, preferring to cross check within their own domain and to look up the source themselves, "And with topics that I already know, in my domains where I know my way around, I can of course immediately cross-check myself whether the whole thing makes sense. And then I would have to cross-check every source it refers to again. Then I might as well look up the source myself." (Interview NU03)[9] Where users draw on their professional knowledge to verify outputs, these non-users name the gap in that same knowledge as where the risk of relying on the output lies. Users rely on their professional knowledge to find mistakes made by LLMs.

[5] „wenn es um Zahlen geht, Wirtschaftsprüfungsberichten zum Beispiel, […] ich überprüfe mich auch selbst doppelt und dreifach und ein Kollege überprüft mich auch nochmal, bevor irgendwas nach außen geht. Und das überträgt sich dann auch auf die Nutzung von KI."

[6] „bei den Projektberichten prüfe ich vor allen Dingen erstmal, ob die Fakten stimmen."

[7] „wenn es wirklich wichtig oder es überrascht mich das Ergebnis, dann schaue ich natürlich nochmal nach."

[8] „Wenn es halt in den Bereich läuft, wo ich dann kein Hintergrundwissen habe und es nicht abschätzen kann, muss ich darauf vertrauen, dass es stimmt, was es mir dort ausgibt. Und da sehe ich so ein bisschen die Gefahr, dass ich dann auf falsche Grundlagen vertraue."

[9] „Und bei, klar, bei Themen, die ich bereits weiß, bei meinen Domainen, wo ich mich auskenne, da kann ich natürlich selber gleich gegenchecken, ob das Ganze Sinn ergibt. Und dann müsste ich ja sozusagen jede Quelle, die es angeht, nochmal gegenchecken. Dann kann ich die Quelle ja eigentlich auch gleich selber raussuchen."

Non-users describe the opposite case, worrying about the areas where they do not know enough to reliably check LLM outputs.

Self-Directed Appropriation of LLM Competence

Almost every user said that they started from a position of doubt and tried LLMs in an open ended way before they could actually see a use for them. IP01 initially saw LLMs as merely a replacement for Google and could not imagine how it would fit and improve their own work, “At first it was more of an experiment, and I was also a bit unsure, like, I do not know if I can really use this for my actual work, because I used to think my work was kind of a closed off world, I mean my job itself, so I used to think it just would not work.” (Interview IP01)[10] IP02 tried LLMs for their master’s thesis and did not find them helpful at the time, “I tried to use it in my master’s thesis. It did not really work out.” (Interview IP02)[11] Even IP03, the most proficient user in this sample, recalled that the early models, while impressive, did not really work for them, and that they only started using one properly later, “There were already tools out there before that, and I did use some of them, […] like DeepL, which was already pretty good back then […] But I think it was still pretty clear to me back then that it did not really work that well. And I probably did not really start using it properly until 2024 with ChatGPT.” (Interview IP03)[12] Across these accounts the users started from doubt and open ended trying and they only found a use for LLMs in the workplace by trying them out themselves.

Using LLMs well was a skill that had to be built up over time and prompting was part of it. In fact, users described effective prompting as a skill that grew through repeated, iteratively improved practice. IP04 stated plainly that prompting has to be learned, “Prompting is a process that you really have to learn well. It takes practice.” (Interview IP04)[13] IP03 made the iterative character of this practice explicit, “It is simply so important to optimize prompts iteratively.” (Interview IP03)[14] IP01 described the same iterating more implicitly, reformulating with more detail whenever an output did not fit until they reached

[10] „das war am Anfang eher so ein Ausprobieren und auch so ein bisschen so nach dem Motto, ah, weiß ich nicht, ob ich das jetzt so wirklich auch für meine Arbeit an sich verwenden kann, weil ich halt auch mal dachte, meine Arbeit wäre so ein geschlossener Raum irgendwie – also mein Job an sich – dass ich halt auch schon mal dachte, das funktioniert gar nicht.“

[11] „Ich habe versucht, das in meiner Masterarbeit zu nutzen. Das hat weniger geklappt.“

[12] „Da gab es ja schon vorher Sachen und ich hab die ja teils auch verwendet, […] [wie] DeepL war damals ja auch schon sehr gut […] Aber ich glaube damals war mir noch sowas von klar, dass das nicht so richtig gut funktioniert. Und ich habe es dann auch tatsächlich so richtig angefangen zu nutzen wahrscheinlich echt erst 2024 mit ChatGPT.“

[13] „Das Prompten ist ein Prozess, den man wirklich gut lernen muss. Das braucht Übung.“

[14] „Es ist halt so wichtig, iterativ Prompts zu optimieren.“

their goal, “Then I just have to be a little more specific or whatever, so I can somehow get there.” (Interview IP01)[15] IP02 traced this practiced skill back to their own experimenting with what works and what does not, “Through all this experimenting, figuring out what works and what does not, it has built itself up.” (Interview IP02)[16] Taken together, these accounts describe a prompting skill that grew through repeated trial and error, with each iteration building on the last.

Users were already familiar with LLMs before using them at work, because they started using them in the private sphere. IP01 first made conscious use of an LLM for a part-time qualification before deploying LLMs at work, “I was pursuing another qualification in part-time alongside my job […] which required me to study, and that was the first time I really made a conscious effort to use AI to gather information and work with it.” (Interview IP01)[17] IP02 described a similar transfer of experience they had already built up in the private sphere, which is what helped them to deploy LLMs at work, “For work it helped me in the sense that I already had a certain body of experience.” (Interview IP02)[18] In both accounts the body of experience the user draws on at work came from their own earlier effort, which they then carried over into their professional use.

By contrast, where this own initiative and curiosity are absent, people never start using LLMs on their own. NU03 described the opposite of curiosity driven exploration, missing the curiosity to try it out, “I do see it in my private environment, whether my husband or my son, who do use something like that and also think it is good […] I am missing a bit of the curiosity to try it out. It is really a felt thing.” (Interview NU03)[19] NU01 knew what learning to use LLMs would take, but was not willing to do it alone at work. They wanted the organization to offer training first, “Just like that, out of nowhere, I might not take it on myself to do that in a work context. But as I already said, if there were some kind of training that shows how to quickly identify certain errors, then I would actually have no problem with it.” (Interview NU01)[20] Where the users figured LLMs out on their own and had the curiosity

[15] „Dann muss ich halt nochmal ein bisschen konkreter formulieren oder so, um halt irgendwie dann ans Ziel zu kommen.“
[16] „Durch dieses ganze Experimentieren, was geht, was geht nicht, hat sich das schon aufgebaut.“
[17] „Ich habe nebenberuflich noch eine Qualifikation gemacht […] dafür musste ich lernen und dafür habe ich das erste Mal so richtig bewusst eine KI verwendet, um mir da Infos zu holen und damit zusammenzuarbeiten.“
[18] „Für die Arbeit hat mir das halt dahingehend geholfen, dass ich halt schon einen gewissen Erfahrungsschatz hatte.“
[19] „Ich erlebe es ja im privaten Umfeld, sei es mein Mann, sei es mein Sohn, die doch so etwas nutzen und das auch gut finden […] Mir fehlt ein bisschen die Neugier, um das auszuprobieren. Es ist wirklch so eine gefühlte Sache.“
[20] „Jetzt so einfach so heraus würde ich mir das im Arbeitskontext vielleicht nicht zutrauen. Aber wie bereits gesagt, wenn es da so eine gewisse Schulung gäbe, die eben zeigt, wie man gewisse Fehler schnell identifizieren

to try LLMs out, these non-users either lacked the curiosity to begin or waited for the organization to start the process for them, so the acquisition never really began.

Task-by-Task Decisions to Deploy or Withhold from Own Judgment

None of the users described an organizational rule, guideline or permitted-task list defining which tasks LLMs should be used for or what information could be entered into them. In the absence of such boundaries, the users in this sample reported that they do not use LLMs for everything but decide task by task from their own judgment. Two users described distinct logics for doing so. IP05, who works in a German federal ministry, found that for emails the effort of prompting outweighed simply writing the email, “But at some point I realized that spending five minutes tweaking this email is just as ineffective as if I had just written it myself.” (Interview IP05)[21] For other work IP05 described starting from the LLM output and checking quickly whether it is somewhere to build on or whether it simply costs too much time, “So, like I said, an LLM first bias. That is right, exactly. And then just check it pretty quickly, so okay, is it somehow halfway there, where I can start from and improve it. Or do I notice that it just costs too much time.” (Interview IP05)[22] IP04 decided by a different logic and drew the line at the phase of the work, using LLMs in the early and middle phases but not for finalization, “For final drafts or finalizing products, rather not. Because then I double-check that what I am doing, what I want to stand behind in the end, is truly adapted to the organizational cultures or language.” (Interview IP04)[23] These are two separate ways of deciding rather than one calculation. IP05 weighs cost against benefit anew for each task, while IP04 draws a line by the phase of the work and keeps the final product to their own judgment. What the two accounts share is that the decision is made by the user from their own understanding of the task.

By contrast, the non-users assessed suitability in broad terms rather than task by task, although their reasons for this varied. NU02 made a blanket assessment that LLMs do not make them faster at all, “It does not make me any faster, I cannot get the task done any

kann, dann hätte ich da eigentlich kein Problem mit.“

[21] „Habe dann aber irgendwann gemerkt, der Fakt, dass ich jetzt hier 5 Minuten an dieser E-Mail rumfeile, ist so ineffective, als wenn ich es einfach mal kurz selber schreibe.“

[22] „Also wie gesagt, LLM first Bias. Stimmt, genau. Und dann halt relativ schnell checken, so okay, ist das irgendwie halbwegs da, wo ich ansetzen kann und es verbessern kann. Oder merke ich, das zieht einfach zu viel Zeit.“

[23] „Für so Final Drafts oder Finalisierung von Produkten eher nicht. Weil da gucke ich dann nochmal, dass es halt an die organisatorischen Kulturen bzw. Sprache… wirklich angepasst ist, was ich mache, was ich vertreten will am Ende.“

quicker if I were to use ChatGPT or some other AI model." (Interview NU02)[24] NU03 reasoned at the same broad level, holding that because the output has to be checked again anyway the tool adds a step rather than saving time, "Because I have to check it again anyway. Then it does not make sense to use it, so to speak." (Interview NU03)[25] NU01 judged at the level of their field as a whole rather than for individual tasks, holding that the work they are assigned does not lend itself to it, "That may also have a bit to do with the task […] I work in HR, also in the area of qualification, so in that respect the task does not really lend itself to it." (Interview NU01)[26] Their specific reasons varied, but none of them identified a single task where the assistance would be worth it. Where users decide task by task, these non-users assess suitability in broad terms, so the task level judgment that characterized the users is absent. The same own judgment also surfaces, more narrowly, when a user draws a boundary of their own. As a brief illustration, IP01 named personal data as an absolute red line they set themselves, "I would never enter personal data… for me that is an absolute red line." (Interview IP01)[27]

[24] „Es macht mich nicht schneller, also ich kann die Aufgabe dann nicht schneller erledigen, wenn ich jetzt irgendwie ChatGPT oder so nutzen würde oder ein anderes KI-Modell."
[25] „Weil ich das eh nochmal überprüfen muss. Dann macht es ja keinen Sinn, das dann zu nutzen, sozusagen."
[26] „Das hängt dann vielleicht so ein bisschen mit der Aufgabe auch zusammen […] ich arbeite ja im Personalbereich, also auch im Qualifizierungsbereich, entsprechend gibt das dann dahingehend die Aufgabe gar nicht mal so her."
[27] „Ich würde niemals personenbezogene Daten eingeben… das ist für mich so ein absolut rotes Tuch."

Discussion

This chapter discusses the findings of both the quantitative and qualitative finding of this thesis. The overarching finding is that LLM adoption in German public administrations did not happen according to the expected organizational route. Rather, it was driven by the individual employee. Instead of waiting for official provision, users gained access, built competence and set the rules themselves, which this thesis terms individualized adoption. The discussion first establishes this finding and the puzzle it poses for the UTAUT framework (6.1). Then, it examines the four functions employees performed themselves to make LLM use possible (6.2). Lastly, it turns turning to what individualized adoption implies for public administrations, namely accountability, data protection and the understanding of technology acceptance in public administration (6.3).

6.1 Adoption ran through the individual and not the organization

Overall, the findings suggest that LLM use and adoption among current users was shaped more by individual initiative than by formal organizational support. Sub-RQ 1 asked which factors predict LLM usage frequency among employees in German public administrations. The regression analysis identified two significant independent predictors, which were Self-Efficacy and Performance Expectancy. Facilitating Conditions, Effort Expectancy and Social Influence did not reach statistical significance, though Social Influence came close. Self-Efficacy captures employees' confidence in their own ability to use LLMs effectively, while Performance Expectancy captures their belief that the tool is useful for their work. In other words, more frequent use was associated with what employees believed they could achieve with the tool and what value they saw in it. Usage frequency was not independently associated no independent association with the organizational conditions surrounding the tools and their users.

This does not mean that organizational provision is irrelevant. Rather, for most users in this sample, organizational provision was simply not the route through which LLM access was made possible. For non-users, however, Facilitating Conditions still seem to matter. Among non-users, FC (Facilitating Conditions) gaps were the most frequently cited barriers: 42.1% reported missing training and guidelines and 36.8% reported missing official access. Organizational support, in other words, did not shape usage among those who had already started, but its absence appears to be what held back those who had not.

This is surprising, because everything about German public administration predicts the opposite. In a formal, rule-bound environment, the factors one would expect to drive professional use of a new technology are organizational, such as official support, technical infrastructure, training, workflow integration or designated points of help. UTAUT argues similarly. In fact, UTAUT gives Facilitating Conditions a direct role in predicting use behavior. Hence, everything in this setting would point to the prediction that the organization or Facilitating Conditions should have mattered most, but it did not.

Based on this finding the second RQ was formulated, which attempted to understand this puzzle: why did individual confidence and perceived usefulness drive usage more than anything the organization provided? The simple answer is that employees did not wait for their organizations to ensure LMM use is possible, but adopted and governed unauthorized LLMs in the workplace themselves. In fact, use grew in a gap between employee demand for LLMs and organizational governance. Those who found LLMs useful and felt able to use them appropriated the tools on their own, reaching for mostly publicly available LLMs and building the competence and judgment they needed through practice. This mixed-methods thesis set out to examine technology acceptance in the first place. But what the findings show is that employees actually face a different challenge, which is access and self-organization. Employees had to find ways to use LLMs on their own, with little formal provision, training or guidance. To do that, they took on parts of the implementation and governance work a public organization would usually take care of. This pattern of employees performing the functions organizational implementation would normally take care of is what this discussion calls individualized adoption.

The rest of this discussion looks more closely at this pattern as well as the implications of individualized adoption for the affected organization. It first looks at the four functions employees performed to make LLM use possible in the workplace, which fall into two categories. The first two concern the resources that use requires, which were gaining access to a tool (6.2.1) and building the competence to use it (6.2.2). The other two concern the which were development of LLM output verification practices (6.2.3) and deciding task by task where LLM use is appropriate (6.2.4). Section 6.3 then turns to what individualized adoption means for accountability and for data protection, for what administrations might do in response and for how technology acceptance is understood in public administration.

6.2 Employees supplied the resources for LLM use

In an ordinary organizational technology deployment, the organization would normally provide both the resources needed to use a tool and the rules for using it. In this sample, both kinds of function were largely shifted to the individual employee. This section begins with the resources, which is access to the tool itself (6.2.1) and the competence to use it (6.2.2). The judgment functions follow in Sections 6.2.3 and 6.2.4, since LLM outputs can appear plausible even when wrong and the tool comes with no built-in boundary for what defines appropriate use.

6.2.1 Access to the tool was supplied by the individual

The first resource employees provided was access to or the LLM tool itself. This is relevant because LLMs differ from many more traditional technologies used in public administrations. A case-management system, database or internal workflow tool usually enters administrative work through procurement, implementation and often necessitates integration into the organizations' IT systems. However, many LLMs are publicly accessible consumer knowledge services. As a consequence, adoption can happen bottom-up and be driven by individual employees rather than by a coordinated organizational rollout.

This pattern is visible in the survey. Among respondents who had used LLMs for work, 83.6% reported using freely accessible tools such as ChatGPT, Perplexity or Claude, while only 13.8% reported using an organization-provided tool such as Microsoft Copilot, LLMoin or NRW.Genius. Therefore, the tool used for administrative work was in most cases not provided through a formal organizational deployment. Employees brought the non-approved tool into their work themselves. In most cases, therefore, the tool used for work-related tasks was a publicly available one that employees accessed themselves rather than one provided by their organization.

This finding helps to understand why Facilitating Conditions did not predict usage frequency in the regression. The Facilitating Conditions items measured organizational support for LLM use, including training or support offerings, workflow integration and contact persons for technical difficulties. However, such support only exists once the organization has introduced a tool in the first place. For most users in this sample that was not the case. Employees could access an LLM without waiting for procurement, training or formal integration. Hence, the absence of organizational provision did not prevent use. Instead, employees resolved the question of access on their own.

The takeaway point is therefore not simply that many users preferred ChatGPT or other public tools. The more important point is that LLM access was decoupled from organizational provision. This decoupling weakens the expected gatekeeping role of UTAUT's Facilitating Conditions. In the theoretical framework, Facilitating Conditions were expected to matter because public administration employees are usually dependent on institutional decisions regarding procurement, infrastructure provision and data protection clearance as already indicated above. Because publicly accessible LLMs removed that dependency, employees could use them without institutional clearance, which is precisely why Facilitating Conditions had no predictive relevance in the regression.

The absence of organizational provision did not prevent access, but it shaped what happened once these external tools were in use. Because employees brought the tool into work themselves, the organization had not provided training, support, quality-assurance routines or clear rules for how to use it. Therefore, individual access created a gap that individuals then attempted to fill themselves.

In sum, the tool-access finding is the first step in the broader argument of this discussion. LLM use became individualized because the first resource needed for use, which was access to the tool, did not have to be supplied by the organization. Once employees could supply access themselves, the organizational adoption process was partially bypassed.

6.2.2 The competence was built by the individual

The second resource employees provided was the competence to use LLMs. When a new technology is introduced through an organizational deployment, the organization does not merely provide access to the tool. Rather, it also creates the conditions under which employees can become competent users through training, introduction and the support that surrounds a new system. In this sample, this competence-building function was largely performed by the individual. Therefore, LLM use became possible because some employees developed the necessary competence themselves. Individual learning was not just part of the adoption process. It actually substituted the organizational implementation function that would have been otherwise not present.

The interviews showed that this competence was not something users already had from the outset. Rather, it was built through the use itself. Users first approached LLMs experimentally and often did not know whether the tool would be useful for their work at all. In several cases, this initial experimental phase happened in the private sphere before it was carried into the workplace. Over time, they learned what kinds of prompts worked, when more specific instructions were needed and how to improve unsatisfactory outputs through repeated interaction. Employees learned by testing what worked, recognizing what failed and adjusting their approach accordingly. This learning pattern confirms the characterization of LLM technology outlined in the context chapter. Prompting is less like a simple technical input and more like a practical technique, one that employees developed not through instruction but through iterative practice.

The survey results are consistent with the idea that competence was built through individual experimentation rather than organizational support. Self-Efficacy was the strongest predictor of usage frequency, while Facilitating Conditions, which included organizational training and support, did not independently predict usage frequency among users. This does not prove that self-directed experimentation caused Self-Efficacy. The data would not allow to make such a causal claim. However, the qualitative findings make an interpretation possible. Employees may have built the Self-Efficacy that predicted more frequent use through their own experimentation, by testing what worked, adjusting their prompts and gradually developing familiarity with the tool.

Since no organizational process guided employees toward competence, only those who took the initiative to experiment with the tool themselves became competent users. In a formal deployment, the organization would normally define what competent use looks like and create a path through which employees acquire that competence. Here, that path was individualized. Some employees became capable users because they experimented, practiced and transferred private experience into the workplace. Others, who lacked curiosity or waited for organizational training, did not begin this learning process. Therefore, the argument here is not simply that users learned how to prompt. It is that competence became unevenly dependent on individual initiative.

6.2.3 Output verification was performed through individualized practices

The first form of judgment concerned the development of verification practices of LLM outputs. This is important because LLM-generated outputs cannot be treated like outputs from more traditional software used in the public administration context. As discussed in the context chapter, LLMs produce probabilistic outputs that may sound plausible even when they are factually wrong, incomplete or unsuitable for the specific administrative context. Therefore, for professional use the relevant question is not only whether employees can get high-quality LLM outputs through effective prompting, but whether they can assess whether those outputs are reliable enough to be used.

In an organizational technology deployment, this verification problem would usually have to be addressed through some form of guideline on quality assurance. The organization could specify which types of outputs require checking, which sources or documents should be used for verification, whether and when a second person should review the result and which outputs should not be used at all. In this sample, however, users did not describe an LLM-specific verification procedure provided by their organization, likely because LLM use had not been introduced through a coordinated organizational process. Instead, each employee developed their own verification practices. They decided individually how to check LLM outputs and whether the results were reliable enough to use. Therefore, responsibility for verification rested with the individual employee rather than with a shared organizational standard.

Users reported two types of verification. The first was an intuitive screening of the LLM's output grounded in professional knowledge. One interviewee described this as a "warning light" going off. Other users spoke of a gut feeling when the output diverged from what they knew about a respective topic. These reactions to LLM output drew on professional experience. Users were able to notice when the model misunderstood a concept, returned implausible information, produced hallucinated data or generated text that that did not fit their working context. Professional knowledge was, in effect, the basis for an initial plausibility check.

The second type was a more deliberate verification type. Users described checking outputs more carefully when the stakes were higher, when numbers or facts were involved, when a document would leave the organization or when an LLM output surprised them. In these cases, users did not merely rely on an intuition. They actively verified whether the output was correct before using it. However, whether something was deliberately checked and how was still left to the individual's discretion. This meant that both the decision of when to verify and the judgment of what to verify rested with the individual user.

The non-user interviews confirm the importance of developing these mechanisms to enable LLM usage in public administration settings from the opposite direction. Non-users did not only object to LLMs because they lacked interest or because the tool seemed difficult. They also worried about situations in which they would not have enough background knowledge to assess whether the output was correct. In such cases, they argued that they would either have to trust the output despite their uncertainty or check every source again themselves, which they considered time inefficient. For these respondents, the need to check the output made LLM use less useful or not useful at all.

This finding also helps to explain why Self-Efficacy was so strong in the quantitative results. Self-Efficacy did not merely capture confidence in operating an interface. It also captured confidence in being able to evaluate the quality and relevance of LLM answers for one’s own professional tasks. The qualitative findings show why this mattered. Frequent use required users to believe that they could control the risk of incorrect output through their own verification practices. This suggests that Self-Efficacy can be interpreted more specifically as confidence in one’s own capacity to judge LLM outputs on top of as general confidence in using the tool. The interviews do not show that users actually eliminated output risks, but they do show that users treated their professional knowledge as the main resource for deciding whether an output was plausible enough to use. Therefore, Self-Efficacy seems to capture the perceived ability to make LLM use professionally manageable under conditions where verification was largely left to the individual. Where this confidence was missing, LLM use became less attractive or was avoided altogether.

The overall takeaway is that responsibility for quality assurance shifted to employees. Users compensated for the absence of formal LLM-specific verification guidance provided by their organization by developing or adapting their own checking practices. These verification practices ranged from informal and ad hoc to more systematic. However, both depended on the individual employee’s professional knowledge and judgment. Therefore, LLM use became possible not because output risks disappeared, but because users took over the task of managing those risks themselves.

6.2.4 Appropriate use was decided task by task

The second form of judgment users exercised concerns deciding where LLM use is and is not appropriate. LLMs do not come with a clear built-in task boundary. Traditional administrative software usually has a self-evident use case, such as that a case-management system manages cases, a database retrieves records and a form system processes predefined inputs. LLMs are different in this regard. They can draft, summarize, translate, brainstorm, rewrite, structure information and assist with many other language-based tasks. Because of this general-purpose character, the question of appropriate use cannot be answered by the character of the tool itself. The user has to actively decide which tasks should be delegated to the LLM and which should not.

As in the previous section, in an organizational deployment, the organization would normally define these boundaries of use through rules, guidelines or at least informal organizational norms. It could specify which tasks are permitted, which data may be entered, which type of outputs require checking and which uses are prohibited. In this sample, however, many employees started using LLMs on their own, rather than as part of a coordinated organizational rollout, which makes it unlikely that their organizations had defined clear boundaries for LLM use.

Both the interviews and a supplementary survey item point in this direction. Of the 18 users who answered the item, 14 reported that their organization had communicated no rules for LLM use or that they were not aware of any (see Section X). Because the item was added later, only a small subsample answered it, so it is treated as corroborating rather than establishing this finding. In sum, the evidence suggests that most users in this sample had no organizational rules which guided their LLM use.

Indeed, the interviews revealed several distinct dimensions along which users exercised this task-level judgment. One was efficiency. Some users assessed whether using an LLM would actually save time or whether prompting, correcting and checking the output would take longer than doing the task themselves. This is important because it shows that users did not treat LLMs as suitable for every task. Instead, they assessed whether the expected benefit justified the effort of prompting, checking and correcting the output. Users made practical calculations about when the tool was worth using. In some cases, they rejected LLM support for simple tasks because the effort of prompting outweighed the benefit of using LLMs.

A second dimension was the phase of work. Some users drew a boundary between early or middle stages of work and final outputs. They used LLMs for drafting, structuring, brainstorming or improving intermediate material, but avoided relying on them for final versions. This boundary is interesting because it shows that users were not only judging task type, but also responsibility. Early-stage LLM support was acceptable because the user could still reshape the result. Finalization required greater ownership, sensitivity to organizational language and confidence that the final product was something the employee could stand behind.

A third dimension was data compliance and sensitivity, especially around personal data. One user described personal data as an absolute red line. This kind of boundary is different from the previous ones. It is not about whether the tool saves time, but about whether a certain type of input should be entered into the tool at all. This shows that users were not only making productivity judgments. They were also making informal compliance judgments in the absence of clear organizational LLM use rules.

Performance Expectancy is especially relevant in this context. The regression showed that Performance Expectancy was the second significant predictor of usage frequency. The qualitative findings suggest that perceived usefulness was not a general belief that “LLMs are useful”. Rather, it was produced through the exercise of task-level judgment. Users perceived LLMs as useful likely because they had learned where the tool created value, where it saved time, where it improved drafts and where it was not worth using. Therefore, Performance Expectancy was not simply a basic belief that LLMs are useful. Rather, it depended on users’ ability to identify the specific tasks and use cases for which LLM use actually improved their work.

In sum, because LLMs have no built-in boundary of appropriate use, employees set that boundary themselves. They weigh effort against benefit, draw lines by work phase and decide what data may be entered at all. Perceived usefulness was likely the product of this task-by-task judgment rather than a general belief that LLMs help.

6.3 Implications

The previous section showed employees taking on four functions the organization would normally take care of. This raises the question of what that shift means in practice for public administrations. Because the sample consisted mostly of frequent users and was not representative, the points below are possibilities the findings point to.

The most important implication concerns accountability. Even when an LLM has helped shape a document, it is the organization, not the individual employee, that remains accountable for the work it produces. Furthermore, according to Bovens (2007), the organization also needs to be answerable for it, which means being able to explain how a respective result came about in the first place (Bovens, 2007).

However, the interviewed users used LLMs early in their work and mostly through freely available tools (83.6% of users) outside the organization's own systems. As a result, the organization has no record of how the result was produced. This is not an issue about whether a mistake has been made. This is first and foremost a data and information problem. Accountability depends on being able to see what happened (Busuioc, 2021). In fact, examples of guidelines for AI in government see keeping a trail of what went in and came out as a basic requirement (e.g., U.S. Government Accountability Office, 2021).

Individualized adoption produces the opposite. It resembles what is known as "shadow IT," where tools used outside approved systems stay invisible to the organization that still bears the consequences (Silic & Back, 2014). This does not mean the interviewed users were careless. In fact, many interviewees described careful and knowledge-based output verification practices. However, because these practices were entirely controlled by the individual and undocumented, the organization has no way to verify that verification actually took place. As a result, accountability remains dependent on individual diligence rather than being embedded in organizational processes.

This loss of sight is most problematic in data protection. Because most users ran on free consumer tools, the data-protection checks that normally come with official procurement were skipped. No user mentioned the existence of organizational rules about what could be entered and the only limit anyone mentioned was one they set for themselves, such as the interviewee IP01 saying entering personal data an absolute red line. It is also still legally and technically unclear how data-protection law applies to these tools at all (Feretzakis et al., 2025). This thesis has no data on whether data was entered into these tools, which were not permissible. Nonetheless, the point here is that this leaves an unmanaged risk within German public administrations, since whether impermissible sensitive data was entered into the tools depended on where each individual chose to draw the line.

The fact that the handling of LLM use is governed by the individual may have even further consequences. For instance, because each employee decides for themselves how carefully to check the tool's output and defines their own task boundary, similar citizen cases may receive very different treatments. Consider the case of a caseworker who consults an LLM to assess whether a specific upskilling measure requested by an unemployed citizen is reasonable and should be approved. Depending on the individual caseworker, the LLM's recommendation may be carefully checked, only briefly glanced at or simply taken over as the LLM suggests. Hence, standardization in how citizen cases are handled can be weakened.

… (Bin nicht Fertig geworden. Hier braucht es noch 2-3 weitere Punkte)

RQs:
Overarching RQ: How does the adoption of Large Language Models take place among employees in German public administrations?

1.1: Which factors predict the frequency of LLM use among employees in German public administrations?

1.2: How do employees in German public administrations make LLM use possible in the absence of formal organizational provision?

Ideen für Implikationen (inkl. Buzzwords):

1. Data protection and confidentiality (als separaten Punkt neben o.g. accountability)
    a. D.h. sensitive personal data is transmitted to third party providers, retained or used for their model training creating breaches of data protection law that public org has never authorized etc.
2. Inconsistency, quality and equity of outputs
    a. D.h. without guidance, training, agreed standards, individuals will use tools in differen ways with no shared verification or quality control process, i.e. uneven output quality risking the – critical in public sector – issue of

potential unequal treatment of citizens bec output for you as citizen will depend on which official you dealt with and how and which tool they used

3. Public trust and legitimacy
   a. D.h. trust in institutions will be undermined if unequal treatment and gov employees just prompting..